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Elastic Link Enables Structural Compliance and Intrinsic Contact Sensing in Robotic Legs

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Abstract

Elastic element in robotic legs offers benefits in terms of adaptability, energy efficiency and robustness. This paper presents a novel concept of robotic leg with a fully elastic arc-shaped link, which serves as both a structural component for motion transmission and a leaf spring. This unique design endows the robotic leg with structural compliance, particularly advantageous for impact load absorption during interaction with the environment. Additionally, the deformation of the elastic link provides the potential for intrinsic contact sensing. A compact deflection sensor, capable of measuring large-scale deformation, is developed and integrated with the elastic link. A contact sensing model is derived using a discretization-based approach to predict the foot contact force and position by leveraging elastic deformation and hip motor torque. To validate the proposed design, a prototype was constructed, and a series of experiments were conducted. The results demonstrate a 35% reduction in impact load compared to the rigid-version robotic leg. Moreover, the mean error of contact force sensing is 1.06 N, and useful terrain information can be obtained during interaction.

Keywords: Robotic leg, elastic link, structural compliance, deflection sensor, contact force sensing.

1. Introduction

Legged robots have the potential to navigate challenging terrains and replace humans in performing dangerous tasks [1, 2]. In recent decades, the field of dynamic legged locomotion has seen significant advancements, due to progress in mechanism design, actuation, and control algorithms. The mechanical design and control of legged robots are often closely interconnected, and skillful mechanical design can alleviate control burdens and lead to improved performance [3].

In the realm of legged robot mechanism design, it is common practice to employ stiff structures to minimize deformation and enhance precision. However, despite these efforts, robotic systems with rigid designs still struggle to replicate the dynamic locomotion capabilities of biological systems. Animals demonstrate remarkable agility and locomotive efficiency with minimal control effort and energy expenditure, thanks to the inherent compliance in their muscles and tendons [4]. Drawing inspiration from this biological advantage, the integration of passive elastic elements into the mechanical design of legged robots can lead to significant improvements in robustness, energy efficiency, and the ability to perform explosive movements.

Compliant actuation systems can be categorized into two primary types based on their configuration: series-elastic actuation (SEA) [5–7] and parallel-elastic actuation (PEA) [8–10]. Tension and torsional springs are frequently employed as the compliant elements within these systems. Spröwitz A. et al. integrated diagonal springs into the pantograph leg design of the quadruped robot Cheetah-cube, which endowed the robot with self-stabilizing capabilities under open-loop control [11]. In another application, a parallel spring embedded within a hollow linkage facilitated powerful jumping maneuvers in low-gravity environments, as demonstrated by the Spacebok robot [12]. The energetic

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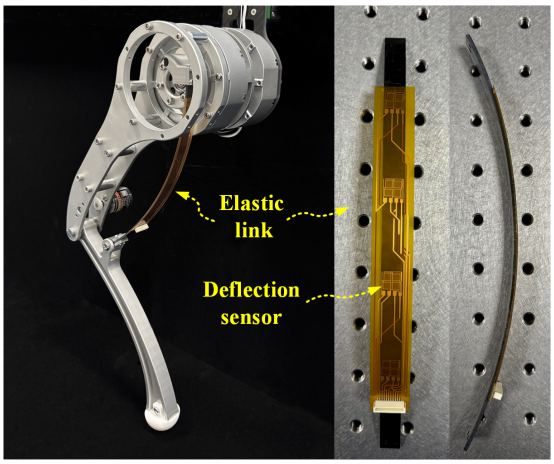


Figure 1: Elastic leg with integrated elastic link and deflection sensors

benefits of parallel elastic actuation have been validated in legged robots, as evidenced by studies such as [13–15]. The straight leaf spring has also been utilized in the design of series-elastic actuation for legged robots. The ATRIAS robot [16, 17], for instance, employs series-elastic plate springs, which enables it to replicate human-like ground reaction forces in spring-mass walking and achieve dynamically stable stepping in 3D. Cassie [18], built upon the ATRIAS design, utilizes SEA plate springs in knee and heel, setting the record for the fastest 100-meter run by a bipedal robot. With these compliant actuation methods, the performance gap between legged robots and biological is narrowing, encouraging more innovative designs in elastic actuation.

Beyond the design of compliant legs, contact force sensing is equally vital for the dynamic locomotion of legged robots. The most straightforward approach is to embed a force transducer directly into the foot [19]. Others have integrated tactile [20], optical sensors [21], micro cameras [22] into specially designed feet with soft pads to achieve contact force sensing. While these direct measurement methods are intuitive and convenient, they often lead to a bulky structural design. Consequently, indirect force sensing methods that avoid placing sensors on the foot have been explored. For instance, using quasi-direct actuators, MIT's mini Cheetah [23] has achieved contact force sensing through motor torques and leg Jacobian. Zhang et al. [24] proposed an external force estimation method for quadruped robots based on generalized momentum . Others have used Kalman filter[25] or extended state observer [26] for contact force estimation of robotic manipulators. However, these sensor-less force estimation methods are generally tailored for rigid-link mechanisms and rely on rigid-body dynamics. Similarly, the Jacobian mapping technique is typically designed for leg designs without inherent compliance, making it unsuitable for direct application to elastic robotic legs.

This work introduces a novel concept for an elastic robotic leg design that utilizes an entire linkage as the compliant element. As depicted in Fig. 1, the proposed elastic leg functions as a unique four-bar mechanism featuring an elastic linkage. Unlike the aforementioned designs, the arc-shaped elastic linkage serves dual purposes: it enables motion transmission as a structural component and also acts as a spring. Due to the elastic linkage design, the leg possesses additional degree of freedom (DOF) for absorbing impact loads, even when the actuators are locked. However, the nonlinear behavior of the leaf spring presents challenges for kinematics analysis and control. Building on our previous research on the large deflection analysis of spatial flexible rods [27], we have derived a kinetostatic model for the elastic leg. Taking into account the significant deformation of the elastic linkage, we propose a method to calculate the foot contact force in both vertical and horizontal directions by leveraging the local deformation of the elastic linkage and the hip motor torque. To measure the local deformation of the elastic linkages, we have designed and implemented specialized gauge sensors suitable for large deformations, as illustrated in Fig. 1. A prototype of the elastic leg has been developed, and its compliant properties and intrinsic contact sensing capabilities have been experimentally validated.

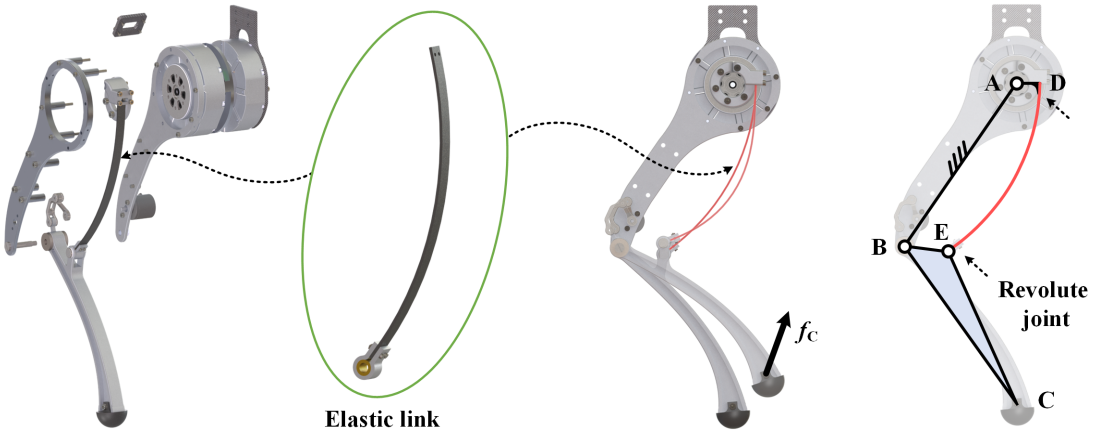


Figure 2: Mechanical design of the proposed elastic robotic leg.

2. Mechanical Design

2.1. Leg design

As shown in Fig. 2, the robotic leg described in this paper has two DOFs, corresponding to hip flexion/extension (HFE) and knee flexion/extension (KFE). To minimize inertia, the hip and knee motors (Unitree GO-M8010-6) are coaxially mounted at the hip. The hip joint is directly actuated by the hip motor, whereas the knee joint is actuated by the knee motor via an elastic four-bar linkage.

It should be noted that the elastic four-bar mechanism designed here differs from traditional rigid four-bar mechanisms in two ways. Firstly, the entire intermediate linkage is elastic, made from 65Mn spring steel and specifically designed in a curved shape to make better the use of its elasticity and avoid buckling. Secondly, the proximal end of the elastic linkage is rigidly connected to the crank, not through a revolute joint, and the distal end connects to the shank link via a passive revolute joint. Consequently, the elastic linkage should not be simplistically considered as a two-force element, which will be particularly paid attention to during modeling in Section 3.

Due to the unique design of the elastic four-bar mechanism, the proposed elastic robotic leg is endowed with compliance when interacting with the environment. This physical compliance is particularly beneficial for impact absorption, helping to enhance the leg's adaptability and stability during locomotion while also safeguarding the leg structure from damage. Furthermore, this physical compliance is independent of the control algorithm, ensuring a sensitive and stable immediate response to external loads during interaction with environment.

2.2. Sensors and Control System

The elastic deformation of the flexible link provides impact load absorption, which can also be used for contact force sensing. To measure the local deflection of the elastic link in real-time, a specifically devised strain sensing system is applied. As demonstrated in Fig. 3, a piece of flexible printed circuit board (FPCB) with four evenly distributed etched strain gauges is attached to the elastic link. A corresponding sampling circuit is designed to acquire the sensing data at a high frequency up to over 1 kHz, which ensures the rapid response of the whole leg system. Further, the sampling system connects the FPCB through the FPC connector and is compact and easy to integrate with a size of $28 \times 36 \times 12$ mm^3. The proposed sensing system provides the leg structure with sensing capability readily.

As illustrated in Fig. 3, the control system for the proposed elastic leg operates on the ROS2 framework. The motor integrates the driver, allowing communication with the host computer via a high-speed RS-485 serial interface. Additionally, an external encoder is employed at the knee joint to measure the joint angle. This measurement is obtained using a parallelogram mechanism, with the acquired angle data solely used for leg parameter calibration.

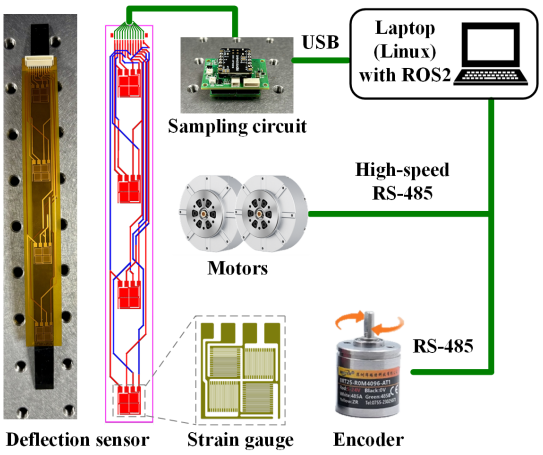


Figure 3: Diagram of the control system.

3. Kinetostatic and Force Sensing Modeling

This section presents the kinetostatic modeling and force sensing methodology of the studied robotic leg. As illustrated in Section 2, the proposed leg comprises of a special four-bar mechanism with an entire elastic linkage. The motion of the point foot is determined by the actuator inputs, external and internal load-induced deformation of the elastic linkage. A discretization-based method, proposed in previous work [27, 28], was used to model the continuous elastic deformation of the link.

The elastic link is discretized into n small segments, and each segment is approximated by a six-DOFs serial manipulator, which is based on the principal decomposition of the structure compliance matrix [29]. For the spring steel stripe used in the studied mechanism, its compression/elongation deformation is negligible compared to bending and torsional ones. Moreover, in the planar case, only the bending effect needs to be considered. Therefore, only one revolute joint is included in the simplified equivalent mechanism.

3.1. Forward and Inverse Problems of Kinetostaics

To control the foot trajectory of the studied robotic leg, the forward and inverse kinetostatic models must first be built. In the forward problem of kinetostatics, the inputs of hip and knee motor angle, namely $q_0,\ q_1 \in \mathbb{R}$, and external force at the foot, namely $\boldsymbol{f}_C = [f_{Cx},\ f_{Cy}] \in \mathbb{R}^{2\times 1}$, are known variables. To determine the equilibrium state for the given inputs, both geometric constraint of the closed-chain mechanism, and force equilibrium in joint space should be taken into account as

$$\begin{cases} \boldsymbol{y} = \boldsymbol{p}_{ABE} - \boldsymbol{p}_{ADE} = \boldsymbol{0} \\ \boldsymbol{\tau} = \mathbf{K}_{\theta}\boldsymbol{\theta} - \mathbf{J}^T \boldsymbol{F}_E = \boldsymbol{0} \end{cases} \tag{1}$$

where $\boldsymbol{p}_{ABE},\ \boldsymbol{p}_{ADE} \in \mathbb{R}^{3\times 1}$ denote the position vectors of E point extracted from the pose matrix $\mathbf{g}_{E1},\ \mathbf{g}_{E2} \in SE(3)$ that are calculated from path 1: "A $\rightarrow$ B $\rightarrow$ E" and path 2: "A $\rightarrow$ D $\rightarrow$ E", respectively, as shown in Fig. 4.

$$\begin{cases} \mathbf{g}_{E1} = \begin{bmatrix} \mathbf{R}_{E1} & \boldsymbol{p}_{ABE} \\ \boldsymbol{0} & 1 \end{bmatrix} = \exp(\hat{\boldsymbol{\zeta}}_0 q_0)\exp(\hat{\boldsymbol{\zeta}}_B \theta_B)\,\mathbf{g}_{E10} \\ \mathbf{g}_{E2} = \begin{bmatrix} \mathbf{R}_{E2} & \boldsymbol{p}_{ADE} \\ \boldsymbol{0} & 1 \end{bmatrix} = \prod_{i=0}^{1} \exp(\hat{\boldsymbol{\zeta}}_i q_i) \prod_{i=1}^{n} \exp(\hat{\boldsymbol{\xi}}_i \theta_i)\,\mathbf{g}_{E20} \end{cases} \tag{2}$$

Here $\hat{\boldsymbol{\zeta}}_0,\ \hat{\boldsymbol{\zeta}}_1,\ \hat{\boldsymbol{\zeta}}_B,\ \hat{\boldsymbol{\xi}}_i \in se(3)(i = 1,\cdots,n)$ are the twists (in 4×4 standard representation) of the hip joint, knee joint, and approximated joints discretized from the elastic link in the home configuration. $\theta_B \in \mathbb{R},\ \boldsymbol{\theta} = [\theta_1,\cdots,\theta_n]^T \in \mathbb{R}^{n\times 1}$ relates to the displacement vector of the corresponding joints. $\mathbf{g}_{E1},\ \mathbf{g}_{E2} \in SE(3)$ represent the pose of the frame attached to revolute joint E from path 1 and 2. $\mathbf{g}_{E10},\ \mathbf{g}_{E20} \in SE(3)$ denote the initial one when $q_0 = q_1 =$

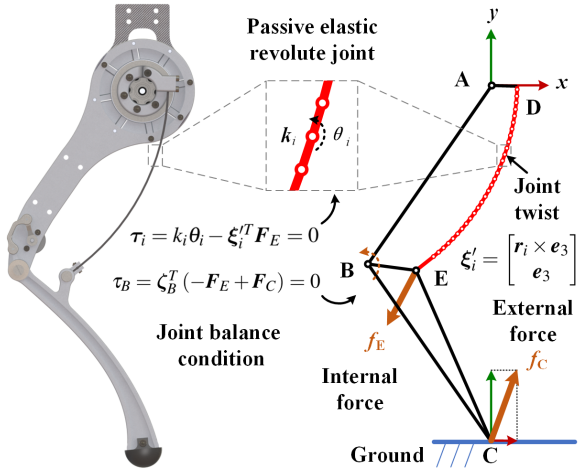


Figure 4: Kinetostatic modeling of the elastic leg.

0, $\theta_B = \pi/2$, $\boldsymbol{\theta} = \boldsymbol{0}$. $\mathbf{K}_\theta = \mathrm{diag}(k_1, \cdots, k_n) \in \mathbb{R}^{n\times n}$ is the diagonal stiffness matrix of the passive elastic joints in the approximated mechanism. $\mathbf{J} = [\boldsymbol{\xi}_1', \cdots, \boldsymbol{\xi}_n'] \in \mathbb{R}^{6\times n}$ is the Jacobian matrix that relates the n-dimensional joint space to the workspace, where $\boldsymbol{\xi}_i' \in \mathbb{R}^{6\times 1}(i = 1, \cdots, n)$ represents the joint twist, in 6×1 vector form, in the mechanism's current configuration. Here, $\boldsymbol{F}_E \in \mathbb{R}^{6\times 1}$ denotes the internal wrench acted at point E, which can be obtained as

$$\begin{cases} \boldsymbol{F}_E = \begin{bmatrix} \hat{\boldsymbol{p}}_{ADE} \begin{bmatrix} \boldsymbol{e}_1 & \boldsymbol{e}_2 \end{bmatrix} \\ \begin{bmatrix} \boldsymbol{e}_1 & \boldsymbol{e}_2 \end{bmatrix} \end{bmatrix} \boldsymbol{f}_E = \mathbf{Q}_E \boldsymbol{f}_E \\ \boldsymbol{F}_C = \begin{bmatrix} \hat{\boldsymbol{p}}_C \begin{bmatrix} \boldsymbol{e}_1 & \boldsymbol{e}_2 \end{bmatrix} \\ \begin{bmatrix} \boldsymbol{e}_1 & \boldsymbol{e}_2 \end{bmatrix} \end{bmatrix} \boldsymbol{f}_C = \mathbf{Q}_C \boldsymbol{f}_C \end{cases} \tag{3}$$

where $\boldsymbol{F}_C$ represents the external wrench applied at the foot, derived by the 2-dimensional force vector $\boldsymbol{f}_C = [f_{Cx},\ f_{Cy}] \in \mathbb{R}^{2\times 1}$. $\boldsymbol{e}_1 = [1,\ 0,\ 0]$, $\boldsymbol{e}_2 = [0,\ 1,\ 0] \in \mathbb{R}^{3\times 1}$ are the unit vectors.

Note that the knee joint is a passive joint, which means the net torque around the knee joint is zero, in the state of equilibrium, namely $\tau_B = \boldsymbol{\zeta}_B^T(-\boldsymbol{F}_E + \boldsymbol{F}_C) = 0$. Therefore, the equilibrium condition for the forward problem of kinetostatis of the studied robotic leg can be represented as a set of nonlinear equations as

$$\boldsymbol{C}_{fwd}(\boldsymbol{\theta},\ \theta_B, \boldsymbol{f}_E) = \begin{bmatrix} \boldsymbol{y} \\ \boldsymbol{\tau} \\ \tau_B \end{bmatrix}_{(n+4)\times 1} = \boldsymbol{0} \tag{4}$$

The forward kinetostatic model relates to a set of $n+4$ nonlinear algebraic equations in terms of $n+3$ unknown variables, namely $\boldsymbol{\theta} \in \mathbb{R}^{n\times 1}$, $\theta_B \in \mathbb{R}$, $\boldsymbol{f}_E \in \mathbb{R}^{2\times 1}$. It should be note that the $\boldsymbol{p}_{ABE}$ and $\boldsymbol{p}_{ADE}$ are 3D position vector, which can degenerate to 2D vector in the studied planar case. Hence, the number of valid equations is $n+3$, equal to that of the unknowns, which implies the forward kinetostatic model is deterministic. Here we used Newton-Raphson method to solve the nonlinear equations (4) efficiently, by making full use of the full system gradient matrix in analytical form $\boldsymbol{\nabla}_{fwd} \in \mathbb{R}^{(n+4)\times(n+3)}$. which can be derived concisely as

$$\boldsymbol{\nabla}_{fwd} = \begin{bmatrix} \dfrac{\partial \boldsymbol{y}}{\partial \boldsymbol{\theta}} & \dfrac{\partial \boldsymbol{y}}{\partial \theta_B} & \mathbf{0}_{3\times 2} \\ \dfrac{\partial \boldsymbol{\tau}}{\partial \boldsymbol{\theta}} & \mathbf{0}_{n\times 1} & \dfrac{\partial \boldsymbol{\tau}}{\partial \boldsymbol{f}_E} \\ \dfrac{\partial \tau_B}{\partial \boldsymbol{\theta}} & \dfrac{\partial \tau_B}{\partial \theta_B} & \dfrac{\partial \tau_B}{\partial \boldsymbol{f}_E} \end{bmatrix}_{(n+4)\times(n+3)} \tag{5}$$

Since the position vector $\boldsymbol{p}_{ABE}$ and $\boldsymbol{p}_{ADE}$ degenerate to 2D, the third row the gradient matrix can be eliminated to obtain a square matrix of $(n+3)\times(n+3)$ size. On the basis of the gradient matrix, the equilibrium condition of forward problem can be determined in an iterative way using the Newton-Raphson method.

$$\begin{bmatrix} \boldsymbol{\theta} \\ \theta_B \\ \boldsymbol{f}_E \end{bmatrix}_{k+1} = \begin{bmatrix} \boldsymbol{\theta} \\ \theta_B \\ \boldsymbol{f}_E \end{bmatrix}_{k} - (\boldsymbol{\nabla}_{fwd})_k^{-1} (\boldsymbol{C}_{fwd})_k \tag{6}$$

For the inverse problem of kinetostatic of the studied robotic leg, the target position of foot and external force applied at the foot are known. To determine the corresponding motor angle q_0, q_1 and equilibrium state of the leg, similar analysis procedure like forward problem can be conducted. The inverse kinetostatic model can be formulated as a set of nonlinear algebraic equation nearly the same as Eq. (4), except for the unknowns.

$$\boldsymbol{C}_{inv}(\boldsymbol{\theta},\, q_1, \boldsymbol{f}_E) = \begin{bmatrix} \boldsymbol{y} \\ \boldsymbol{\tau} \\ \tau_B \end{bmatrix}_{(n+4)\times 1} = \boldsymbol{0} \tag{7}$$

Note that the knee joint angle θ_B and hip motor angle q_0 are not counted as the unknown variables in Eq. (7), because these two joints angle can be directly calculated according to simple geometric relation of the 2-R mechanism for a given target foot position. The system gradient matrix of inverse kinetostatic model $\boldsymbol{\nabla}_{inv}$ can also be derived analytically, and the equilibrium configuration and motor inputs of the inverse problem can be determined iteratively.

$$\boldsymbol{\nabla}_{inv} = \begin{bmatrix} \dfrac{\partial \boldsymbol{y}}{\partial \boldsymbol{\theta}} & \dfrac{\partial \boldsymbol{y}}{\partial q_1} & \boldsymbol{0}_{3\times 2} \\ \dfrac{\partial \boldsymbol{\tau}}{\partial \boldsymbol{\theta}} & \boldsymbol{0}_{n\times 1} & \dfrac{\partial \boldsymbol{\tau}}{\partial \boldsymbol{f}_E} \\ \dfrac{\partial \tau_B}{\partial \boldsymbol{\theta}} & \dfrac{\partial \tau_B}{\partial q_1} & \dfrac{\partial \tau_B}{\partial \boldsymbol{f}_E} \end{bmatrix}_{(n+4)\times(n+3)} \tag{8}$$

Similarly, the equilibrium configuration and motor inputs of the inverse problem can be determined iteratively. in the following update theme as

$$\begin{bmatrix} \boldsymbol{\theta} \\ q_1 \\ \boldsymbol{f}_E \end{bmatrix}_{k+1} = \begin{bmatrix} \boldsymbol{\theta} \\ q_1 \\ \boldsymbol{f}_E \end{bmatrix}_{k} - (\boldsymbol{\nabla}_{inv})_k^{-1} (\boldsymbol{C}_{inv})_k \tag{9}$$

3.2. Local-Deformation-Based Contact Force Sensing

As shown in Fig. 2, four strain gauges are deployed at one side of the elastic link. And their sensor readings are approximately proportional to their bending angle. This property can be leveraged to realize contact force sensing for the proposed elastic leg on the basis of kinetostatic model, given the motor angles and deflection sensor readings.

The bending angle of a deflection sensor simply corresponds to the sum of joint displacement it covers [30, 31]. Thus, the local bending deformation gathered by these sensors can be formulated as an extra constraint as

$$\boldsymbol{s} = \mathbf{S}_e\boldsymbol{\theta} - \boldsymbol{s}_0 = \boldsymbol{0} \tag{10}$$

where $\boldsymbol{s}_0 = [s_1, s_2, s_3, s_4] \in \mathbb{R}^{4\times 1}$ represents the deflection sensor readings, and $\mathbf{S}_e \in \mathbb{R}^{4\times n}$ is a constant selection matrix relating the position of deflection sensors, each row of which is given by

$$\mathbf{S}_{e,i} = [0,\cdots,1,\cdots,1,0,\cdots,0] \in \mathbb{R}^{1\times n},\ i = 1,2,3,4 \tag{11}$$

whose element one corresponds to the passive elastic joint covered by the deflection sensors. Theoretically, one gauge is enough to determine the external force, here we select four to enhance the robustness.

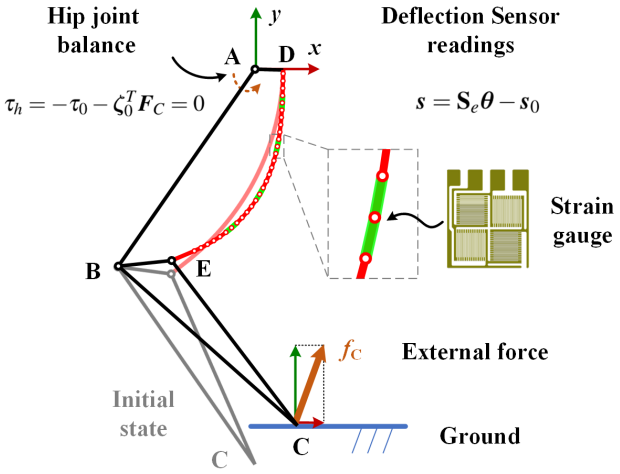


Figure 5: Contact force sensing modeling of the elastic leg.

With Eq. (4) and (10), the external force's component tangential to shank link can be solved, since the equation in (4) only provides a torque equilibrium constraint. In order to sense the complete external force applied at the foot, the torque equilibrium at hip was added as one another constraint as

$$\tau_h = -\tau_0 - \zeta_0^T F_C = 0 \tag{12}$$

where τ_0 represents the hip motor torque, which can be obtained directly from the motor status. Combining Eq. (4), (10) and (12), the foot contact force sensing model can be established as follows

$$C_{sns}(\theta,\ \theta_B, f_E, f_C) = \begin{bmatrix} y \\ \tau \\ \tau_B \\ s \\ \tau_h \end{bmatrix}_{(n+9)\times 1} = \mathbf{0} \tag{13}$$

Following a similar way as ∇_{fwd} and ∇_{inv} , the gradient matrix of Eq. (13), namely ∇_{sns}, can be derived conveniently, since most of the component can be reused. Then, the gradient-based search algorithm can be used to solve the nonlinear equations efficiently through the following update theme as

$$\begin{bmatrix} \theta \\ \theta_B \\ f_E \\ f_C \end{bmatrix}_{k+1} = \begin{bmatrix} \theta \\ \theta_B \\ f_E \\ f_C \end{bmatrix}_{k} - (\nabla_{sns})_k^+ (C_{sns})_k \tag{14}$$

where $(\nabla_{sns})_k^+$ is the pseudoinverse of gradient matrix ∇_{sns} to (13). It should be noted that not only the foot contact force can be sensed, but the shape of the elastic leg can be simultaneously reconstructed. This means that the entire leg configuration, including the foot position, can be obtained at the same time.

4. Experimental Validation

In this section, a variety of experiments are conducted to verify the effectiveness of the design, modeling and contact sensing method. First, a foot trajectory experiment is performed to verify the accuracy of the kinetostatic model for the robotic leg with an elastic link. Then, the inherent structural compliance of the design is demonstrated

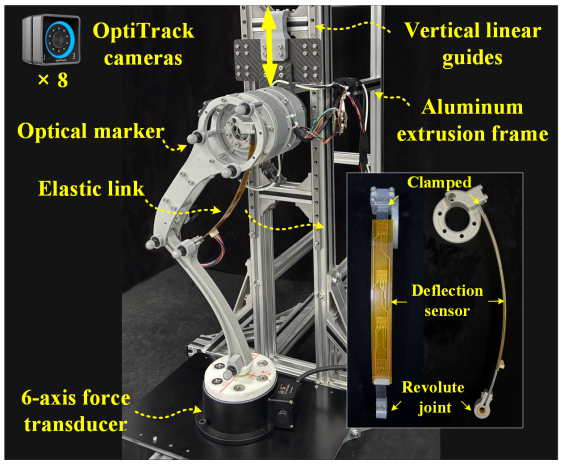


Figure 6: Experimental setup for validating the structural compliance and intrinsic contact sensing capabilities of the elastic leg.

through comparative hardware tests. Quantitative contact force sensing experiments are conducted to assess the feasibility of the proposed force sensing method. Finally, qualitative terrain classification experiments are carried out to demonstrate the potential of the intrinsic contact sensing capability for terrain identification.

4.1. Experimental Setup

To conduct the validation experiments, a prototype of the proposed elastic leg was developed, along with a custom-built apparatus. The apparatus features an aluminum frame with vertical linear guides and a commercial six-axis force transducer. As depicted in Fig. 6, the robotic leg is mounted on the slides, allowing vertical movement and the option to lock the slides to the aluminum frame. The force transducer is mounted on the ground plate to measure the contact force between the foot and the ground. Optical markers are attached to the thigh and shank links to capture the leg's movement using the OptiTrack system. For comparison, a rigid carbon fiber link is used in the free-fall tests to replace the elastic link.

4.2. Free-Fall Tests

To validate the benefits of structural compliance introduced by the elastic link design, free-fall tests were conducted. In these experiments, the robotic leg was lifted to a height of 100 mm and then released freely to contact with the ground. The contact force was measured using a force transducer throughout the experiment. For comparison, the

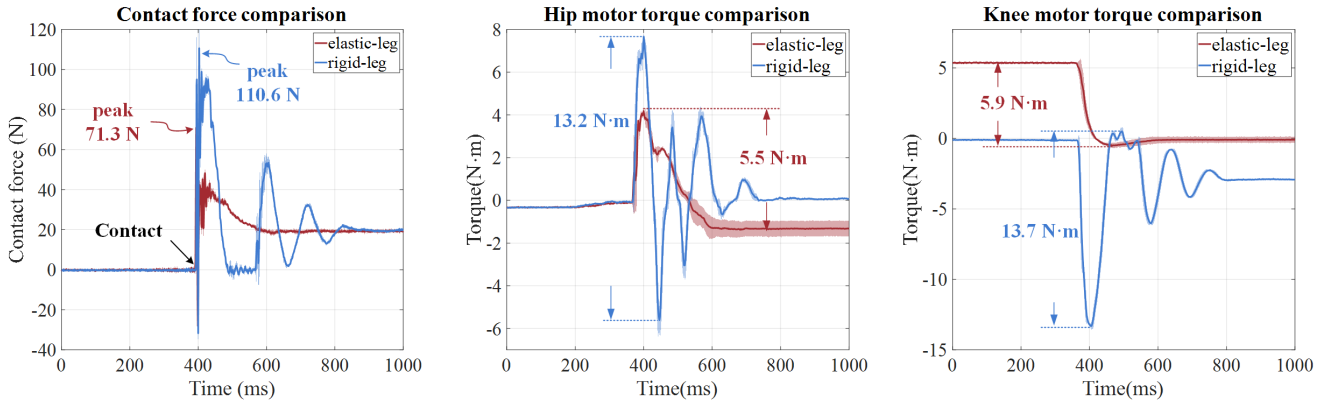


Figure 7: Comparison of contact forces and motor torques for the elastic leg and rigid leg upon free-fall release. The experiment was repeated five times, with the red and blue solid lines representing the mean values, and the shaded areas indicating the standard deviations.

Table 1: Force sensing results in varying foot horizontal positions

Force	Error	#1	#2	#3	#4	#5
Vertical	RMS (N)	0.89	0.52	0.91	1.02	1.02
	Max (N)	2.16	1.25	1.79	2.38	2.41
	Relative	3.6%	1.8%	3.5%	4.6%	3.0%
Horizontal	RMS (N)	0.77	0.52	1.06	0.95	0.68
	Max (N)	1.96	1.07	2.33	1.93	1.25
	Relative	26.9%	17.5%	22.4%	26.0%	50.5%

same procedure was carried out with a rigid-link version of the robotic leg. The free-fall tests were conducted five times for both the elastic and rigid legs.

As shown in Fig. 7, a significant impact load is observed at the moment of foot-ground contact. For the rigid link leg, the maximum impact load reached 110 N (the self-weight of the robotic leg is about 1.8 kg). Moreover, the contact force for the rigid leg exhibited substantial oscillations, taking approximately 400 ms to stabilize. In contrast, the elastic leg experienced an impact load of 71 N, representing a 35% reduction compared to the rigid link, and the data indicated a quicker stabilization time of around 200 ms. This demonstrates the superior performance of the elastic leg design in mitigating impact loads upon ground contact. From the perspective of motor torque, significant impacts to hip and knee motor can also be observed as shown in Fig. 7. The difference between the maximum and minimum torque of the motor during the contact process is considered the impact load on the motor in this experiment. As a comparison, the impact loads on the hip and knee motors are 5.5 $\mathrm{N}\cdot\mathrm{m}$ and 5.9 $\mathrm{N}\cdot\mathrm{m}$, which is much smaller than those of the rigid leg (13.2 $\mathrm{N}\cdot\mathrm{m}$, 13.7 $\mathrm{N}\cdot\mathrm{m}$), resulting a significant reduction of 58.3% and 56.9%. These results highlight the advantages of the proposed elastic link design in reducing impact loads on the robotic leg.

4.3. Intrinsic Contact Sensing

A. Contact Force and Position Sensing in Static Scenarios

Contact force and foot position sensing experiments were conducted to validate the intrinsic contact sensing capabilities of the proposed elastic leg. In the force sensing experiments, the hip of the leg is secured to an aluminum frame, with the robotic foot in contact with the ground. A variety of target foot positions are set to achieve varying levels of contact, resulting in different magnitudes of contact forces. Specifically, five distinct horizontal target positions (from -50 mm to 50 mm) are chosen, and for each horizontal position, seven vertical positions are set with an increment of 1 mm. At each horizontal position, the foot contact with the ground gradually intensifies, and this experiment was repeated five times. The sensed contact forces, calculated by the sensing model, were compared with the measured ones by the force transducer. The error of the sensed forces, in the five selected horizontal positions, are shown in Table 1. The Root Mean Square (RMS) and maximum error of the sensed vertical contact forces are similar to those of the horizontal forces, with an RMS of approximately 1 N and a maximum error of 2.4 N. The relative error for the sensed vertical contact forces is below 5%, with a maximal vertical force of 35 N. Because the maximum horizontal force is small (around 3 N), the relative error for the sensed horizontal contact forces is relatively large, but the absolute error of both horizontal and vertical forces is close. In addition, the vertical load exerted on the hip can be also be sensed through vertical contact force sensing. As shown in Fig. 8 (a), the hip of the robotic leg is left free and standard weights of 1 kg, 2 kg, 3 kg and 4 kg are attached to the hip, and the estimated results fit the nominal values close. These results highlight the effectiveness of the elastic leg in accurately sensing contact forces across a range of positions.

Apart from contact force, the foot position can also be sensed at the same time, which can be used to estimate the height of the contact objects. As shown in Fig. 8 (b), the robotic leg steps on a simple stair with six steps, each 8 mm in height. The height of the step surfaces can be estimated based on the sensed foot position when the foot makes contact with a step. The sensed step heights are very close to the nominal values, demonstrating accurate estimation of the height of the contact objects.

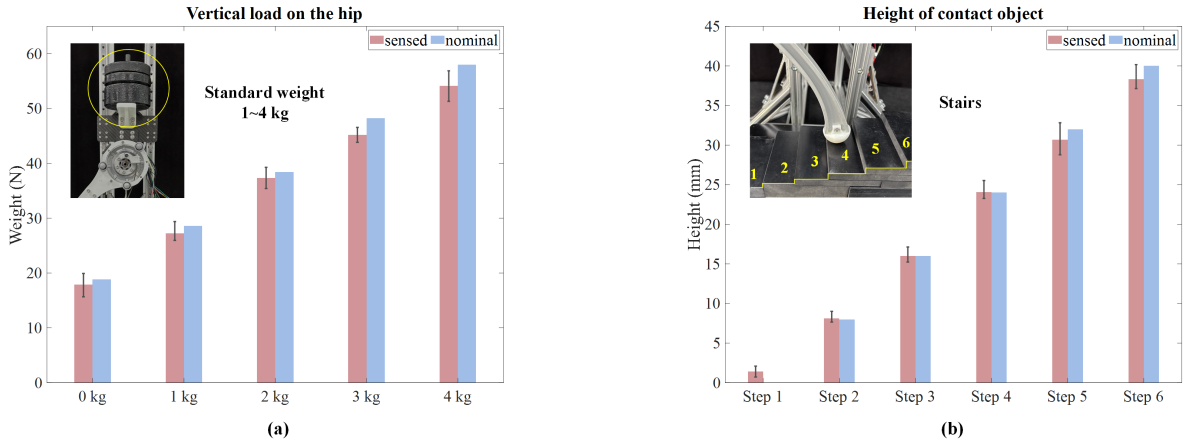


Figure 8: Results of vertical hip load sensing and contact object height sensing. (a) Comparison of the sensed and measured contact force in the horizontal and vertical directions; (b) Comparison of the sensed and nominal step heights. The experiment was repeated five times, with the bars representing the mean/nominal values and the error bars indicating the distribution range of the data across the five trials.

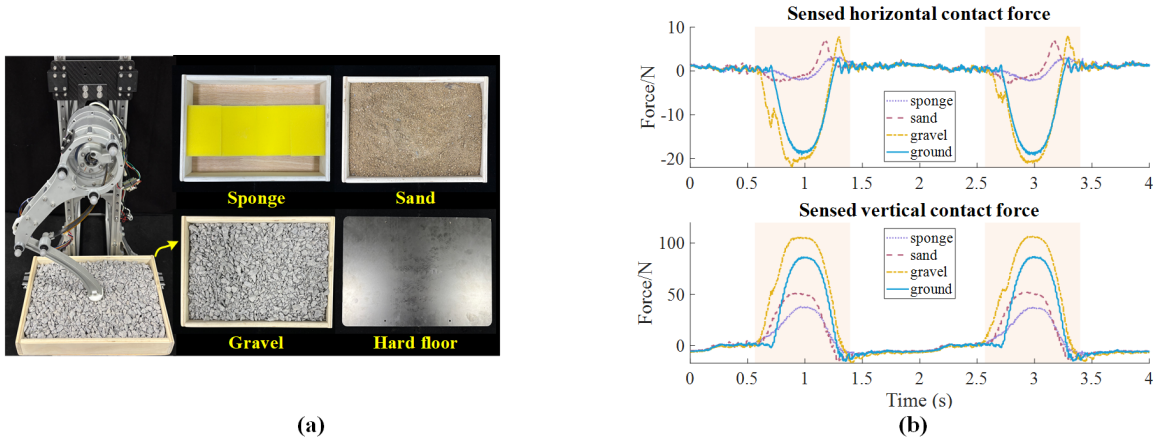


Figure 9: Experimental setup and results of terrain classification. (a) Experimental setup; (b) Sense horizontal and vertical contact forces as the elastic leg steps on different terrains. The shaded area represents the period during which the foot is in contact with the ground surface.

B. Terrain Classification

With the structural compliance and intrinsic contact sensing capability brought by the elastic link design, valuable environmental information during contact can be captured. Terrain classification experiments were conducted to validate its potential for terrain identification. In these experiments, terrains with varying physical properties, including sand, gravel, hard ground, and sponge, were selected, as shown in Fig. 9 (a). The robotic leg stepped vertically and repetitively onto each terrain, while the sensed contact forces are recorded. From the results presented in Fig. 9 (b), distinct differences in the trends of the contact force can be observed. As shown in Fig. 9 (b), from the perspective of contact force, various features emerge during the interaction. Specifically, different peaks and slopes in the contact force curve during the contact period can be observed in the lower part of Fig. 9 (b). The terrain hardness is reflected in the slope of the force curve. For instance, the sponge surface is the softest, exhibiting the smallest slope and peaks in the force curve during contact. In contrast, harder surfaces such as gravel or hard ground have relatively larger slopes and peaks in the force curve. Additionally, when the foot of the elastic leg contacts the gravel surface, relative sliding and bouncing occur between the gravel particles, which is reflected in abrupt changes in the force curve (upper part of Fig. 9 (b)).

Table 2: Terrain classifier results using features extracted from sensed contact forces

		Predicted Class			
		Gravel	Sand	Sponge	Ground
True Class	Gravel	93.8%	0.0%	0.0%	6.2%
	Sand	4.4%	91.2%	4.4%	0.0%
	Sponge	0.0%	8.7%	91.3%	0.0%
	Ground	9.5%	0.0%	0.0%	90.5%

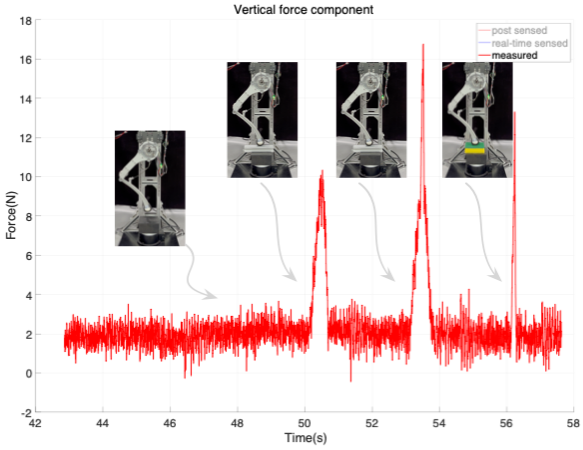


Figure 10: Force sensing results for contact detection.

Benefiting from the property of structural compliance and force sensing capability, useful terrain information can be captured in the recorded data. These characteristic patterns in the recorded data enable the effective distinction of various terrains, making it possible to design classifiers capable of identifying different types of terrain. A simple support vector machine (SVM) classifier was developed. The sensed contact force data were segmented into individual strides to train the classifier, using features extracted from the sensed contact forces. The selected features include peak amplitude, average amplitude, peak width (defined as the width of the reference line at half the peak height), and area under the curve [32]. For each terrain, 40 strides were selected, with 50% of the data used for training and 50% used for validation. As shown in Table 2, the terrains can be effectively distinguished with over 90% accuracy in a single stride.

4.4. Demonstration of Force Sensing and Control in Dynamic Scenarios

Leveraging the integrated foot force sensing capability, force feedback control was implemented on the developed elastic robotic leg. Two scenarios were demonstrated using the prototype: 1) continuous jumping based on contact event detection, and 2) adaptive walking on uneven terrain utilizing contact force control.

A. Contact Event Detection

Accurate detection of foot-terrain interaction, including contact location and force magnitude, is crucial for robotic locomotion. The real-time sensed contact force was employed to detect contact events, as illustrated in Fig. 10. During repeated leg flexion and extension motions, contact with various unexpected objects (e.g., rigid aluminum profiles, hard rubber pads, or soft sponge pads) was reliably identified through the force feedback. To intuitively demonstrate this capability, the leg was programmed to retract immediately upon the contact force exceeding a predefined threshold, as shown in the supplementary video. A key advantage of this approach is that it obviates the requirement for dedicated foot force sensors or binary contact switches, thereby simplifying the robot's structural design.

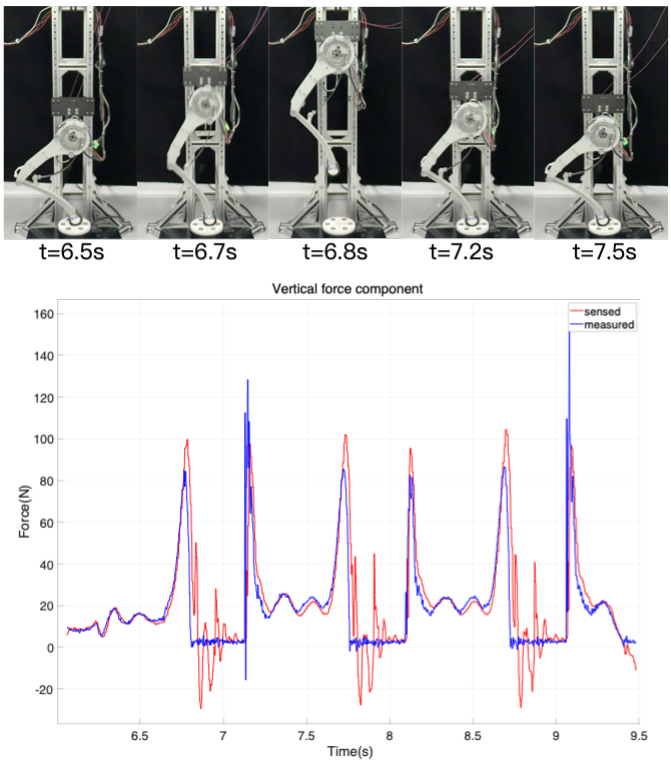


Figure 11: Force sensing results during continous jumping.

B. Continous Jumping

Beyond static contact sensing, force sensing experiments were also conducted in dynamic scenarios. As illustrated in Fig. 11, the robotic leg performed continuous vertical jumping motions while its hip was constrained by a vertical linear guide rail. In this scenario, significant impact forces arise during the brief period immediately following foot-ground collision. Sensing these impact forces is crucial for dynamic environmental interaction. During the jumping cycles, both the contact forces sensed in real-time using the proposed method and those measured by the reference force transducer were recorded (Fig. 11). The contact event detection technique described in Section A was employed for finite state machine transitions, with the corresponding control block diagram shown in Fig. X.

The results reveal a rapid increase in foot contact force just prior to liftoff, attributable to the push-off action. During the flight phase, the measured contact force (from the transducer) is zero, confirming no ground contact. However, the sensed contact force exhibits fluctuations around zero due to elastic vibrations in the elastic link. Upon subsequent ground contact, another impact force peak occurs. As shown in the results, the sensed contact forces (both horizontal and vertical components) closely track the measured reference forces in terms of trend and magnitude. This demonstrates the applicability of the proposed force sensing method to dynamic scenarios involving impact loads.

C. Adaptive Walking on Uneven Terrain

Leveraging the force sensing capability, a force control strategy was designed to regulate foot-ground contact forces during locomotion on unknown, uneven terrain. A comparative experiment evaluated walking performance. The uneven terrain was constructed using randomly positioned objects of varying sizes (e.g., rocks, bricks, grass, sand, rigid rubber pads, aluminum blocks) to simulate surfaces with height variations and differing compliance levels. A six-axis force transducer, mounted on a vertical slider, provided indirect measurement of ground reaction forces during walking trials.

Without Force Control: Only 4 out of 10 trials successfully traversed the simulated environment. Furthermore, contact forces spiked rapidly when the foot encountered elevated obstacles. With Force Control: Upon contact event

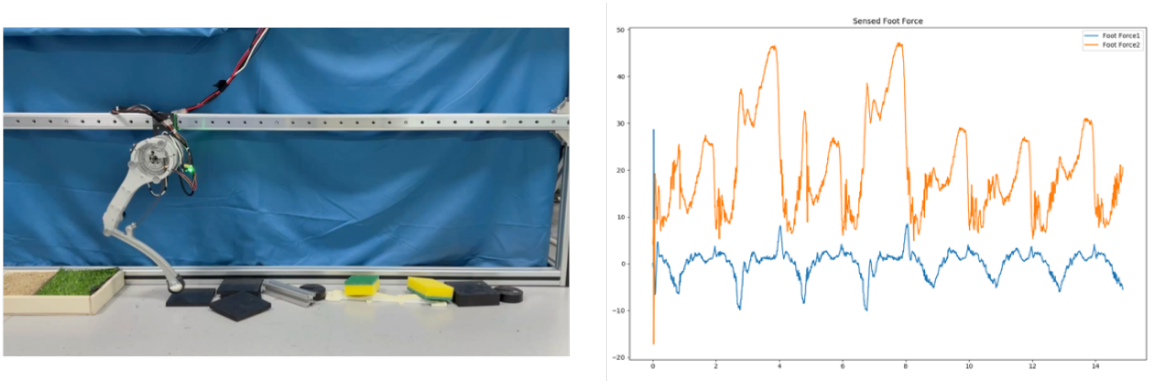


Figure 12: Force sensing results for contact detection.

detection, the contact force was actively regulated below a predefined threshold. This strategy resulted in successful traversal in all 10 trials. As shown in Fig. 12, although a slight force increase occurred when stepping onto higher obstacles, the implemented control successfully maintained the overall contact force below XX N.

5. Conclusions

In this paper, a novel elastic robotic leg with a fully elastic link is proposed. This elastic link, with an arc-shaped design, endows the robotic leg with structural compliance and can also be used for contact force sensing. Taking into account the nonlinear elastic deformation of the link, both a kinetostatic model and a contact force sensing model are derived using a discretization-based method. The elastic deformation of the elastic link upon interaction with the environment can help to mitigate impact loads, enhancing adaptability to terrain in unstructured environments. A specialized deflection sensor, integrated with a FPCB and a corresponding compact sampling circuit, is designed to measure the local deformation of the elastic link. By leveraging the deformation data and hip torque, the foot contact force can be identified in both the vertical and horizontal directions.

A prototype of the proposed elastic leg is developed, and a variety of experiments are carried out to validate the effectiveness of the design and force sensing method. The results show that the elastic leg can precisely follow planned foot trajectories using the kinetostatic model. Additionally, a 35% reduction in impact load was observed in free-fall tests, compared to a rigid-link robotic leg of the same size, validating the effectiveness of the elastic link design in impact mitigation. Furthermore, the contact force sensing accuracy at various contact locations was assessed, with a mean error of 1.06 N. Finally, different terrains were classified based on the characteristic patterns extracted from the sensed contact force. Considering the inherent structural compliance and intrinsic contact sensing capability, the implementation of the proposed robotic leg in quadruped and biped robots will be investigated in future work.

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